The Child Penalty in Consumption: Evidence from Japan

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Abstract

A large literature documents the substantial decline in women's earnings following childbirth. This paper documents a similarly large and persistent penalty in women's consumption expenditures. Using the Japanese Panel Survey of Consumers, which separately records spending by individual household members and common household expenditures, we examine how spending is reallocated following childbirth. After the first birth, women's expenditures decline by 43%, compared with a 58% decline in earnings, and remain substantially lower for at least ten years. In contrast, expenditures assigned to husbands and total household expenditure remain largely unchanged. The wife's share of household expenditures falls by 5.3 percentage points, with larger and more persistent declines among women whose employment does not recover following childbirth. We examine two potential explanations: mothers may sacrifice more of their own consumption because they are more altruistic toward their children, or childbirth may weaken women's bargaining power within marriage. To assess the first explanation, we compare expenditure responses to the first and second births. Unlike the first birth, the second birth is associated with similar declines in women's and men's expenditure shares, suggesting that a simple explanation based on mothers consistently sacrificing more for their children is incomplete. To investigate the bargaining channel, we exploit Japanese pension reforms in 2008 that improved women's financial security following divorce. The estimates reveal no clear effect on intrahousehold expenditure allocation, although they are imprecise.

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1 Introduction

The birth of a child has important implications for a couple's economic well-being. A large literature documents the substantial decline in women's earnings following childbirth (Kleven et al. 2019). However, much less is known about how the economic costs of childbirth are distributed within the household. The arrival of a child creates new consumption needs, requiring couples to allocate resources to an additional household member, often at a time when the mother's earnings are declining. Yet these changes do not necessarily imply that mothers experience a larger decline in consumption than fathers. Couples may pool their incomes and share the economic costs of childbirth, insulating mothers from some of the consequences of their earnings losses. The extent to which they do so determines how the burden of childbirth is distributed between spouses. Understanding how childbirth affects the allocation of consumption within households is therefore essential to assessing its economic consequences for women.

In this paper, we examine how the allocation of consumption between spouses changes following childbirth and investigate the mechanisms underlying these changes. Answering this question is difficult with conventional household expenditure data, which typically record expenditures at the household rather than individual level. Researchers therefore commonly infer individual resource shares by examining how spending on goods consumed by particular household members, such as men's and women's clothing, varies with total household expenditure (Browning et al. 1994; Dunbar et al. 2013; Lechene et al. 2022), or by using equivalence scales. We instead use a unique dataset from Japan that provides detailed information on expenditures assigned to individual household members.

These data come from the Japanese Panel Survey of Consumers (JPSC), an annual longitudinal survey of women that began in 1993. A distinctive feature of the survey is that respondents report their household's monthly expenditures separately across five categories: expenditures for the wife, the husband, children, other household members, and common expenditures. This breakdown allows us to distinguish expenditures assigned to each spouse from those shared by the household. Moreover, because the survey follows the same women over time and collects information on childbirth and both spouses' earnings, we can examine how the allocation of household expenditures evolves around childbirth, alongside changes in each spouse's earnings.

Our analysis reveals a large and persistent decline in women's individual expenditures following childbirth. We first document the well-established child penalty in earnings in our

Japanese sample: women's earnings fall sharply following the birth of their first child, while men's earnings remain largely unchanged. More strikingly, expenditures assigned to women fall to approximately half of their pre-birth level and remain substantially lower even ten years after childbirth. In contrast, men's expenditures show no comparable decline. Despite the addition of a new household member, total household expenditures also remain relatively stable. These findings suggest that households accommodate the new consumption needs associated with childbirth largely by reallocating existing expenditures, with a disproportionate decline in spending assigned to women. Correspondingly, the wife's share of total household expenditures declines substantially alongside her share of couple earnings, even though a decline in one does not mechanically imply a decline in the other.

We next examine how these patterns vary with women's labor-market attachment. Women with higher earnings shares before childbirth tend to have higher expenditure shares in the years following childbirth. Similarly, among women employed before childbirth, those who return to employment in subsequent years have higher expenditure shares than those who do not. The time-use data reveal a similar gender disparity. Following childbirth, women's paid work hours and leisure decline, while their time spent on housework and childcare increases. The corresponding changes for men are substantially smaller. However, husbands contribute more to weekday housework in households where their wives return to employment. Together, these findings suggest that the allocation of expenditures, labor supply, and household responsibilities are closely related following childbirth.

To understand why women's expenditures decline so sharply following childbirth while men's remain largely unchanged, we consider two possible explanations. The first is that mothers are more altruistic toward their children than fathers are, and therefore sacrifice more of their own consumption to provide for their children. If mothers reduce their own consumption to provide for their children, the arrival of a second child could lead to further reductions in their expenditures. However, we find that women's expenditures remain relatively stable following the second birth. While this pattern does not rule out differences in parental altruism, it suggests that differences in parental altruism alone are unlikely to fully explain the observed patterns.

An alternative explanation is that childbirth reduces women's bargaining power within the household, leading to a decline in their share of household resources. To formalize this idea, we develop a simple collective model of the household in which the allocation of resources depends on each spouse's bargaining weight. Childbirth may weaken women's bargaining position by worsening their options outside marriage. For example, mothers may

face greater childcare responsibilities following a divorce, worse marriage-market prospects, and reduced labor-market opportunities. These changes could lower women's utility outside marriage, weakening their bargaining position within the household.

To investigate whether women's outside options affect the allocation of household resources, we exploit a pension reform introduced in Japan in April 2008. The reform gave women who were financially dependent on their husbands the right to receive a share of their husbands' pensions in the event of divorce without requiring their husbands' consent. By improving women's financial circumstances following divorce, the reform may have strengthened their bargaining position within marriage, without directly changing household resources during marriage.

Our empirical analysis proceeds in two steps. First, we examine whether the reforms changed the allocation of expenditures between spouses, comparing households that were differentially exposed to the reforms based on their employment characteristics. Second, we investigate whether the reforms mitigated the decline in women's expenditure shares following childbirth. We compare changes in expenditures around childbirth across birth cohorts with different exposure to the reforms, using a triple-differences strategy that exploits differences in eligibility.

Preliminary results do not reveal a clear change in intrahousehold expenditure allocation following the reforms, either generally or around childbirth. However, the estimates are imprecise, and we cannot rule out economically meaningful effects. In ongoing work, we are exploring additional datasets that would allow us to investigate this mechanism with greater statistical precision.

Distinguishing between these mechanisms has important implications for understanding the economic consequences of childbirth and designing policies to address them. If the decline in women's expenditures reflects a reduction in their bargaining power, policies that improve women's economic opportunities and financial security outside marriage could help mitigate this decline. For example, policies that improve women's financial security in the event of divorce could influence how resources are allocated within marriage, even without directly affecting household income. Thus, understanding the role of bargaining power is important for identifying policies that can protect women's economic well-being following childbirth.

This paper contributes to two strands of literature. First, we contribute to the literature on child penalties, which has primarily focused on the effects of childbirth on women's

employment and earnings (Angelov et al. 2016; Kleven et al. 2019; Andresen and Nix 2022; Kleven et al. 2025); see Cortés and Pan (2023) for a review of this literature. We extend this literature by examining how childbirth affects the allocation of consumption within households by using a unique dataset on individual consumption expenditures. We document a large and persistent decline in women's expenditures following the birth of their first child, with no comparable decline for men. Contemporaneous work by Guo and Xie (2024) and Becker (2026) also documents changes in intrahousehold expenditure allocation following childbirth using Japanese panel data.

Second, we contribute to the literature on intrahousehold bargaining and the role of outside options in determining the allocation of household resources (Rangel 2006; Voena 2015; Lise and Yamada 2019). We exploit a pension reform in Japan that provides plausibly exogenous variation in women's outside options, without directly changing household resources during marriage. This allows us to test whether strengthening women's outside options mitigates the decline in their share of household expenditures following childbirth.

The rest of the paper is organized as follows. The next section provides institutional background on Japan, followed by a description of the JPSC and the analysis sample in Section 3. Section 4 presents the main empirical findings, including changes in expenditures, earnings, and time use following childbirth. Section 5 examines heterogeneity in these patterns, and Section 6 investigates potential mechanisms. Section 7 concludes.

2 Institutional Background

This section provides institutional context for our analysis by describing four features of the Japanese setting: trends in women's employment, patterns of marriage and childbearing, maternity and parental leave policies, and household financial management practices.

Women's Employment. Women's employment in Japan has increased substantially over the period covered by our data. Among women aged 15–64, the employment rate rose from 56.7 percent in 1993 to 71.5 percent in 2021.

Marriage and Childbirth. Marriage and childbearing are closely linked in Japan, where cohabitation and nonmarital births remain uncommon (OECD 2024). In 1993, only 1.2 percent of births occurred outside marriage, increasing to just 2.3 percent by 2021 (National Institute of Population and Social Security Research 2025). This contrasts sharply with the

United States, where the corresponding share increased from 31.0 to 40.0 percent over the same period (Federal Interagency Forum on Child and Family Statistics 2025). Marriage has also been increasingly delayed in Japan, with the average age at first marriage for women rising from 26.1 years in 1993 to 29.5 years in 2021 (National Institute of Population and Social Security Research 2025).

Maternity and Parental Leave Policies. Japan provides both maternity leave and parental leave to eligible workers. Maternity leave consists of six weeks before childbirth and eight weeks after childbirth, with earnings-related benefits available to qualifying workers. In addition, Japan introduced job-protected parental leave in 1992, allowing eligible parents to take leave until their child reaches one year of age. Initially unpaid, parental leave benefits were introduced in 1995 at 25 percent of prior earnings and subsequently increased to 40 percent in 2001 and 50 percent in 2007. Since 2014, eligible parents have received 67 percent of their previous earnings during the first six months of parental leave and 50 percent thereafter (Asai 2015; Yamaguchi 2019).

Although both parents are entitled to parental leave, take-up has historically been much higher among women. In fiscal year 2021, 85.1 percent of female employees took parental leave, compared with only 14.0 percent of male employees (Cabinet Office of Japan 2023).

Household Financial Management. A distinctive feature of Japanese households is the traditional practice of wives managing household finances. Under a common arrangement, husbands transfer their earnings to their wives, who manage household expenditures and provide their husbands with a personal spending allowance, known as *kozukai* (pocket money). Although this practice has become less prevalent over time, wife-managed household finances remain common, particularly among couples with children (Çineli and Mugiyama 2023).

The JPSC collects information on household financial management, including the husband's monthly pocket-money allowance for certain financial-management arrangements. In the appendix, we examine this measure as an additional check on our findings. Our main analysis instead uses expenditures assigned specifically to each spouse, which are recorded separately in the survey.

3 Data

3.1 The Japanese Panel Survey of Consumers

The primary data source is the Japanese Panel Survey of Consumers (JPSC), an annual longitudinal survey of women initiated in 1993 by the Institute for Research on Household Economics and administered by Keio University since 2018. The survey draws from five entry cohorts (1993, 1997, 2003, 2008, 2013) of both married and unmarried women. Respondents in the initial cohort were aged 24–34 at entry, while sbequent cohorts recruited women ages 24–29. The full sample contains 29 survey waves through 2021 and approximately 4,100 women.

The JPSC has two features central to this paper. First, it asks each respondent to decompose total household expenditure in September of each survey year into five categories: (1) expenses shared by the entire family, (2) expenses for the respondent (the wife), (3) expenses for the husband, (4) expenses for children, and (5) expenses for other household members. These expenditures are reported in yen and allow us to construct individual expenditure levels and shares, while distinguishing expenditures assigned to each spouse from those shared by the household.

Second, the JPSC records detailed time-use information for both spouses. For both the wife and the husband, the survey records hours spent on paid work, housekeeping and childcare, hobbies and leisure, and sleep and personal care.

The survey also collects information on the employment status, earnings, and industry of both the respondent and her spouse, as well as changes in household composition, including childbirth and divorce. The longitudinal structure allows us to identify the timing of childbirth and construct event-time measures to examine how household expenditures evolve before and after birth.

3.2 Analysis Sample and Variable Construction

The main analysis focuses on women whose first childbirth is observed in the JPSC. We define event time relative to the first birth, with $t = 0$ denoting the year of childbirth, $t = -1$ the year before childbirth, and $t = 1$ the year after childbirth. Our man analysis covers the period from five years before to eight years after the first birth ($t \in [-5, 8]$).

The analysis sample differs modestly from the full JPSC sample in observable characteristics. Women in the analysis sample married at an average age of 29.5 years, compared

with 28.0 years in the full sample. They also have slightly higher educational attainment, with an average education code of 4.58 compared with 4.02 on a seven-point scale.

Our analysis focuses on two outcomes: (1) expenditures assigned to each spouse and (2) each spouse's share of total household expenditures. We construct expenditure shares by dividing expenditures assigned to each spouse by total household expenditures, defined as the sum of the five expenditure categories described above.

4 The Child Penalty in Earnings, Consumption and Time-Use

This section examines how earnings, household expenditures, and time use evolve following the birth of the first child. We begin by documenting the well-established child penalty in earnings in our Japanese sample: women's earnings decline sharply following childbirth, while men's earnings remain largely unchanged, consistent with the broader child-penalty literature (Kleven et al. 2019). We then apply the same event-study framework to examine how childbirth affects the allocation of expenditures within the household. We document a large and persistent decline in women's individual expenditures and their share of total household expenditures, with no comparable decline in men's individual expenditures. Finally, we examine changes in time use, showing that women reduce their market work and leisure time while increasing time spent on housework and childcare. In contrast, the corresponding changes for men are substantially smaller.

4.1 The child penalty in earnings

Following Kleven et al. (2019), we estimate the following event-study specification separately for each outcome Y:

$$Y_{it} = \sum_{k \neq -1} \beta_k^Y \mathbb{1}\{K_{it} = k\} + \sum_a \alpha_a^Y \mathbb{1}\{A_{it} = a\} + \sum_\tau \gamma_\tau^Y \mathbb{1}\{\text{year}_{it} = \tau\} + \varepsilon_{it}, \tag{1}$$

where K_{it} denotes event time relative to the first birth, A_{it} denotes age, and yearit denotes calendar year. The specification includes age and calendar-year fixed effects to account for life-cycle and aggregate time trends. The year immediately preceding childbirth ($Kit = -1$) is the omitted reference period.

To express the estimated changes in the outcome as percentages, we divide each event-time coefficient by the average predicted outcome at that event time, after removing the estimated event-time effect.

$$P_k^Y = 100 \times \frac{\hat{\beta}_k^Y}{N_k^{-1} \sum_{i,t:K_{it}=k} \left(\hat{Y}_{it} - \hat{\beta}_k^Y \right)}, \tag{2}$$

where N_k is the number of observations at event time k, and $\hat{Y}_{it} - \hat{\beta}_k^Y$ is the predicted outcome excluding the event-time effect.

4.2 Child Penalties in Earnings and Employment

We begin by examining the child penalty in earnings and employment. Figure 1 shows a sharp and persistent decline in women's earnings following the birth of their first child, consistent with the well-established child-penalty literature. The long-run earnings penalty for women is -58%, while men's earnings remain largely unchanged (6%). Women's employment falls sharply at birth (at $t = 0$) and recovers gradually over the following decade (on average over $t \in [5, 10]$), while men's employment does not change ().

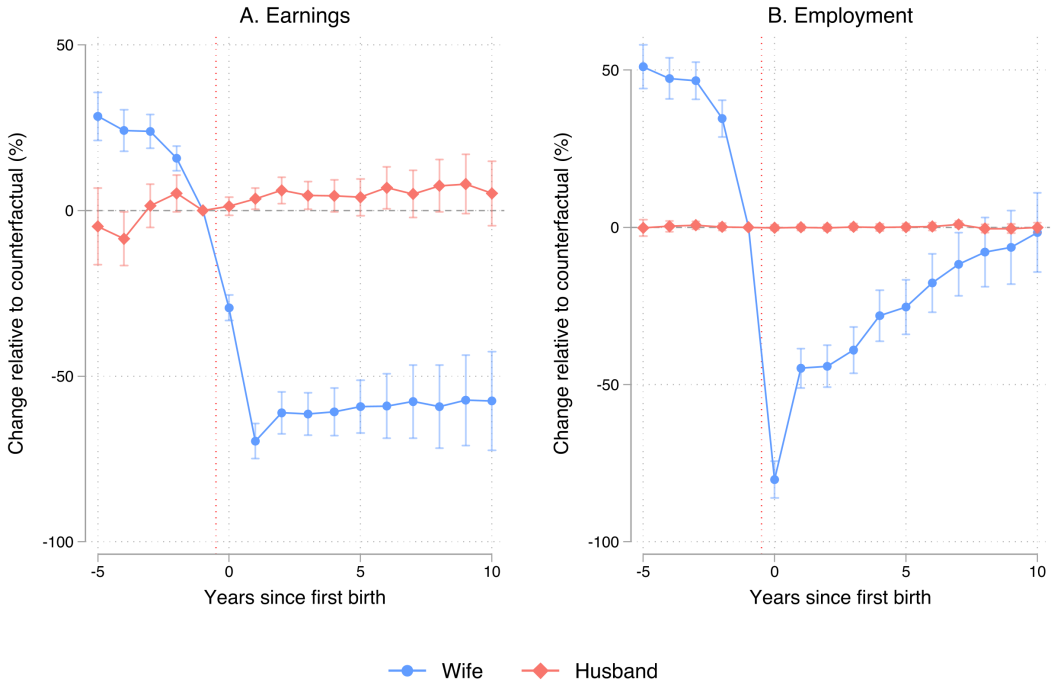


Figure 1: Child penalties in earnings and employment

Notes: Event-study estimates from equation (1): each outcome is regressed on event-time indicators (omitted category $t = -1$), single-year age indicators (the husband's age for husband outcomes), and calendar-year indicators. Coefficients are expressed as a percentage of the counterfactual outcome, as in equation (2). Panel A: annual earnings in real 2015 yen. Panel B: employment, an indicator for currently working. Wives' earnings and employment are already falling in the years before the birth, so the reference year $t = -1$ is below earlier years.

Sample: Women whose first birth is observed in the panel, event times −5 to 10. Bars show 95 percent confidence intervals based on standard errors clustered by woman.

Source: JPSC.

4.3 Child Penalties in Individual Expenditures

We next examine how expenditures assigned to each spouse change following childbirth. The decline in women's earnings does not necessarily imply a corresponding decline in their expenditures, as couples may pool their incomes and share the economic costs of childbirth. However, Figure 1 shows that expenditures assigned to women decline sharply following the birth of their first child and remain substantially lower in the long run. The long-run expenditure penalty for women is -43%, approximately 74% as large as their earnings penalty. In contrast, men's expenditures remain largely unchanged (-7%).

Interestingly, total household expenditures remain relatively stable (4%) despite the addition of a new household member. This suggests that households accommodate the new consumption needs associated with childbirth largely by reallocating existing expenditures, rather than increasing total spending. Our findings indicate that this adjustment is concentrated in expenditures assigned to women, while expenditures assigned to men remain largely unaffected.

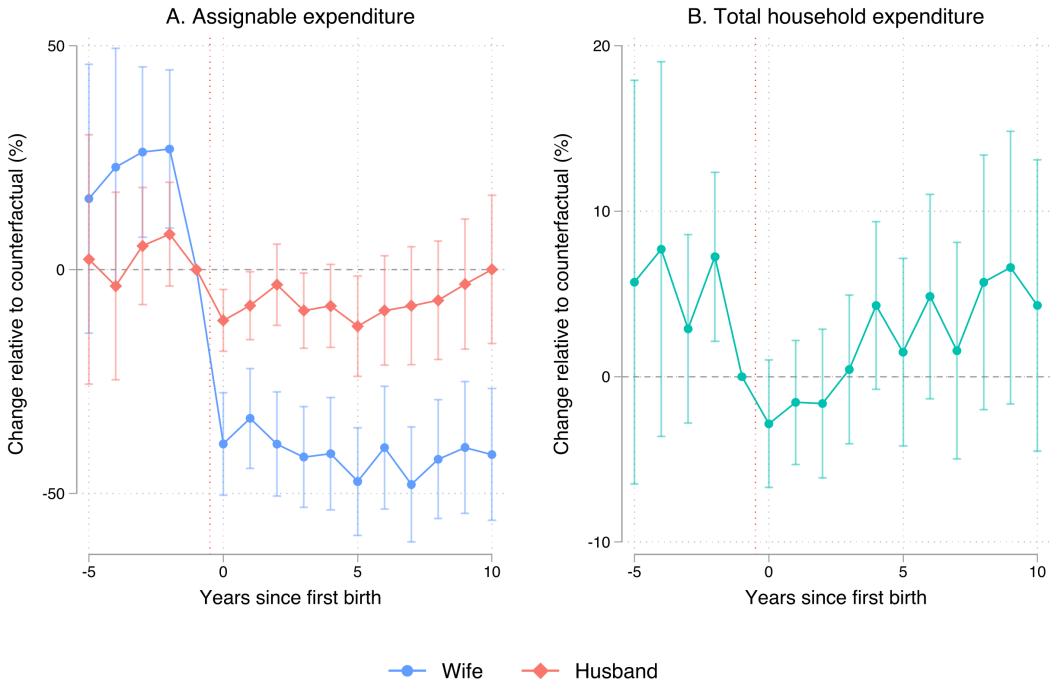


Figure 2: Child penalties in assignable and total household expenditure

Notes: Event-study estimates from equation (1): each outcome is regressed on event-time indicators (omitted category $t = -1$), single-year age indicators (the husband's age for husband outcomes), and calendar-year indicators. Coefficients are expressed as a percentage of the counterfactual outcome, as in equation (2). Panel A: monthly spending assigned to the wife and to the husband. Panel B: total monthly household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses), estimated with the wife's age. Both in real 2015 yen.
Sample: Women whose first birth is observed in the panel, event times -5 to 10. Bars show 95 percent confidence intervals based on standard errors clustered by woman.
Source: JPSC.

4.4 Changes in Intrahousehold Expenditure Shares

We next examine how the distribution of household expenditures changes following childbirth. Figure 3 shows that the wife's share of the couple's total earnings declines by 20.4 percentage points, while her share of total household expenditures declines by 5.3 percentage points. Importantly, the decline in the wife's expenditure share does not mechanically follow from the decline in her earnings share. Couples may pool their resources, allowing them to share the economic consequences of women's earnings losses. However, we find that women's share of household expenditures falls alongside their share of earnings.

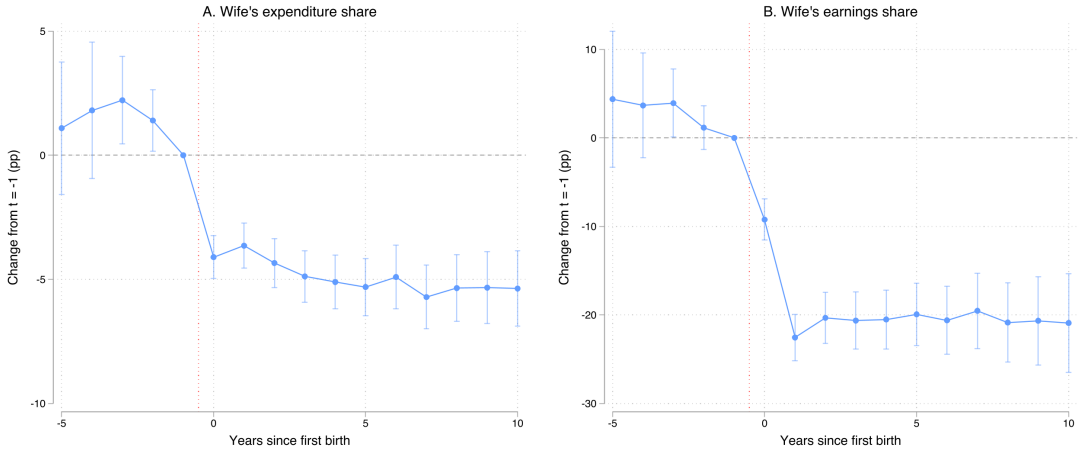


Figure 3: The wife's shares of household expenditure and earnings around first childbirth

Notes: Event-study estimates from equation (1): each outcome is regressed on event-time indicators (omitted category $t = -1$), single-year age indicators (the husband's age for husband outcomes), and calendar-year indicators. Both regressions use the wife's age; coefficients are in percentage points relative to $t = -1$. Panel A: the wife's expenditure share, spending assigned to her divided by total household expenditure. Panel B: the wife's earnings divided by the sum of the wife's, the husband's, and other household members' earnings.

Sample: Women whose first birth is observed in the panel, event times -5 to 10. Bars show 95 percent confidence intervals based on standard errors clustered by woman. Y-axis scales differ across panels.

Source: JPSC.

A decline in the wife's expenditure share could also arise simply because the arrival of a child increases total household expenditures, even if expenditures assigned to the wife remain unchanged. However, as documented in the previous subsection, women's expenditures decline substantially in levels as well. Appendix Figure A1 presents the raw trajectory of the wife's earnings share, while Appendix Figure A2 shows the decomposition of household

expenditure shares, including an increase of 15.1 percentage points in the children's share by $t = 8$.

To further examine these patterns, we estimate a complementary event-study specification for the wife's expenditure share that includes woman fixed effects, with $t = -1$ as the omitted reference period and standard errors clustered at the woman level. Figure 4 presents the results. The pre-birth coefficients are close to zero and statistically insignificant. The wife's expenditure share declines sharply in the year of childbirth and remains persistently lower thereafter. By $t = 8$, the estimated decline is 3.3 percentage points (95% confidence interval: [2.4, 4.1]). These results confirm that the decline in women's expenditure share is also observed within women over time, after accounting for time-invariant individual characteristics.

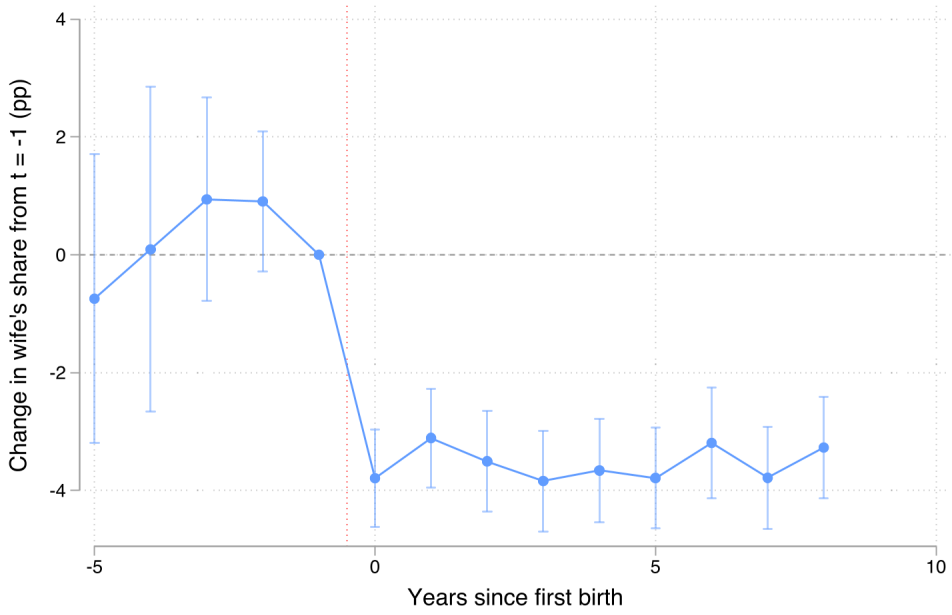


Figure 4: Child penalties in expenditure with individual fixed-effects

Notes: Coefficients from a regression of the wife's expenditure share on event-time indicators ($t = -5, \ldots, 8$; omitted category $t = -1$) with woman fixed effects and no other controls. The outcome is in share units (0.01 = 1 percentage point). The expenditure share is spending assigned to the person divided by total household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses).

Sample: Women whose first birth is observed in the panel, all observations with a valid share at event times −5 to 8. Bars show 95 percent confidence intervals based on standard errors clustered by woman.

Source: JPSC.

4.4.1 Measurement robustness

We examine two potential measurement concerns: changes in the composition of respondents reporting individual expenditures and changes in how expenditures are classified following childbirth.

Reporting Selection. The availability of individual expenditure data varies across event time. Before childbirth, 41.2% of observations contain a valid expenditure decomposition, compared with 91.5% after childbirth. If women who report expenditures differ systematically from those who do not, changes in the composition of respondents could affect the estimated expenditure trajectories.

To address this concern, Figure 5 compares the expenditure-share trajectory in the full unbalanced sample with that of women who report valid expenditure decompositions both before ($t \in [-3, -1]$) and after ($t \in [6, 8]$) childbirth. Both samples exhibit a sharp decline in the wife's expenditure share at childbirth and similar post-birth trajectories. The differences between the two series are concentrated in the pre-birth period.

We further examine the sensitivity of these results to alternative sample restrictions in Figure 6, using the event-study specification with woman fixed effects. Restricting the sample to women with valid expenditure reports in both the pre- and post-birth windows yields an estimated decline of 3.2 percentage points at $t = 8$ (95% CI: [2.3, 4.2]), compared with 3.3 percentage points in the full sample. We also restrict the sample to women who do not have a second child during the event window, to examine whether subsequent births contribute to the long-run expenditure patterns. The estimated decline remains substantial at 2.5 percentage points (95% CI: [0.6, 4.3]).

Finally, we examine whether the results are driven by particular recruitment cohorts. Excluding each of the five JPSC entry cohorts in turn yields estimated declines at $t = 8$ ranging from 2.9 to 3.6 percentage points. Appendix Figure A3 also shows a decline in the wife's expenditure share within each cohort. Together, these exercises suggest that the main findings are not driven by changes in reporting composition, subsequent births, or any single recruitment cohort.

Expenditure Classification. A second concern is that childbirth may change how respondents classify household expenditures. Because the wife reports the expenditure breakdown, some spending previously attributed to her may instead be recorded as spending on children

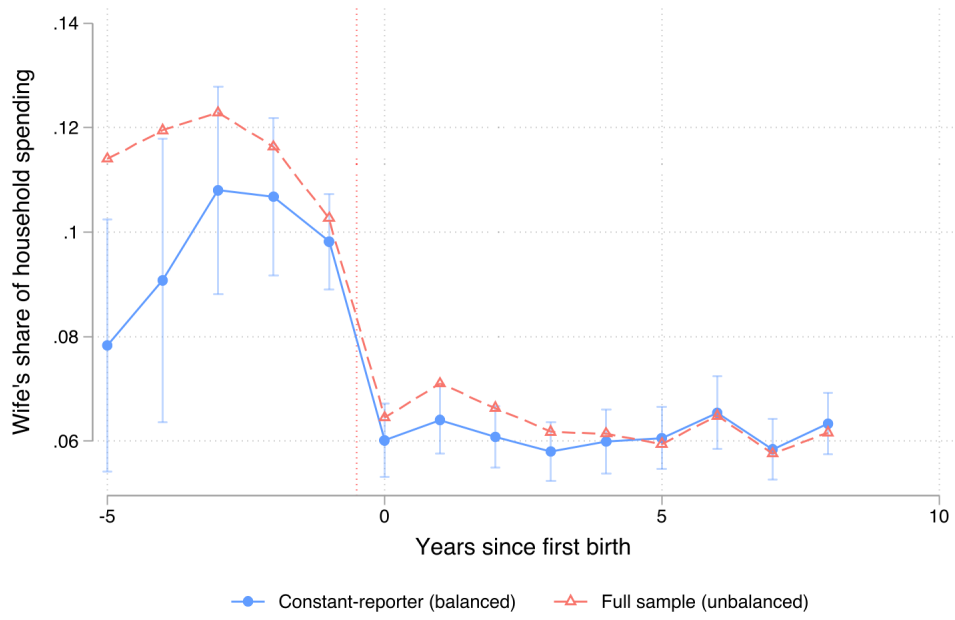


Figure 5: Wife's expenditure share: Full sample and constant reporters

Notes: Mean wife expenditure share by event time, in share units. The expenditure share is spending assigned to the person divided by total household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses).
Full sample: all women whose first birth is observed in the panel. *Constant reporters*: women with at least one valid expenditure share in $t \in [-3, -1]$ and at least one in $t \in [6, 8]$. Bars show 95 percent confidence intervals of the constant-reporter means.
Source: JPSC.

or as common household expenditures following childbirth. This could contribute to the measured decline in her individual expenditures.

We first examine whether the decline in the wife's expenditure share is driven mechanically by the addition of expenditures on children to the household budget. Figure **??** reports the wife's share of expenditures assigned to either spouse, defined as wife/(wife + husband), excluding expenditures on children, other household members, and common household expenses. We find that this share also declines substantially following childbirth. By $t = 8$, the estimated within-woman decline is 3.3 percentage points (95% CI: [0.8, 5.8]). Thus, the decline in the wife's expenditure share is not simply a mechanical consequence of including children's expenditures in the denominator. Moreover, as shown in Figure 7, expenditures assigned to the wife also decline in absolute yen, indicating that the results are not driven solely by changes in total household spending.

As an additional check, we examine the husband's personal spending allowance, known

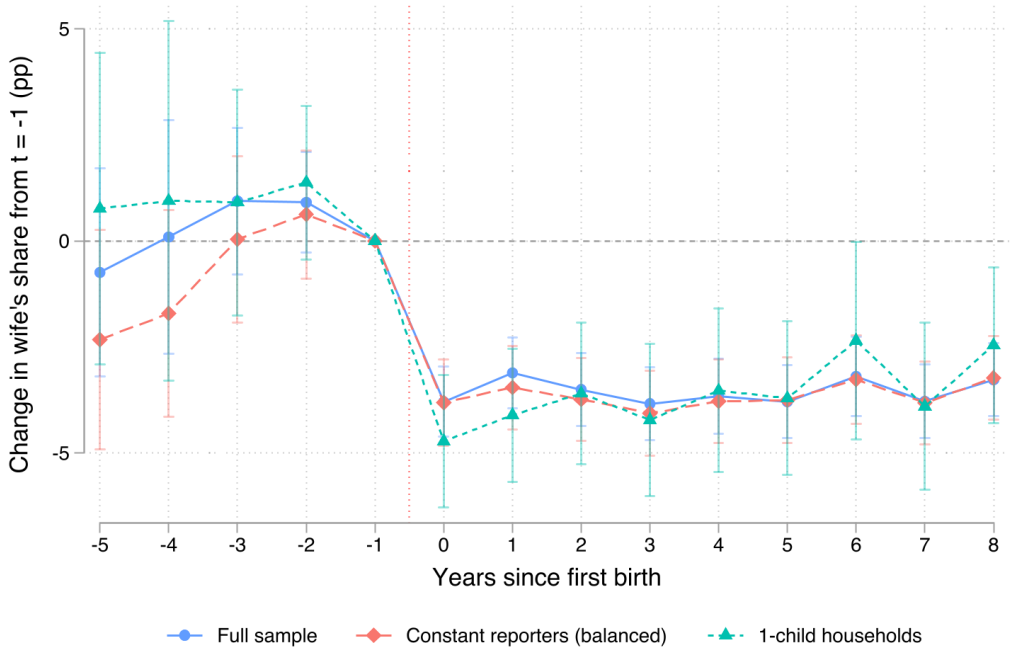


Figure 6: Robustness of expenditure-share estimates to sample restrictions

Notes: Coefficients from the woman fixed-effects event study of Figure 4, in percentage points relative to $t = -1$, estimated on three samples: all women whose first birth is observed in the panel; constant reporters (at least one valid share in $t \in [-3, -1]$ and at least one in $t \in [6, 8]$); and women with no second birth observed in the panel. Event times -5 to 8. Bars show 95 percent confidence intervals based on standard errors clustered by woman.
Source: JPSC.

as *kozukai*, which the JPSC records separately for certain household financial-management arrangements. Appendix Figure A9 shows that the husband's allowance remains relatively stable around childbirth, while Appendix Figure A10 shows little variation across women's post-birth employment groups.

These results show that the decline in the wife's expenditure share is not simply a mechanical consequence of increased spending on children. However, we cannot rule out the possibility that some expenditures previously attributed to the wife are instead classified as children's or common expenditures following childbirth.

4.5 Changes in Time Use

We next examine how the allocation of time between spouses changes following childbirth. Figures 8 show changes in hours spent on paid work, leisure, and housework and childcare on a typical workday. Following the birth of the first child, women's hours in paid work decline

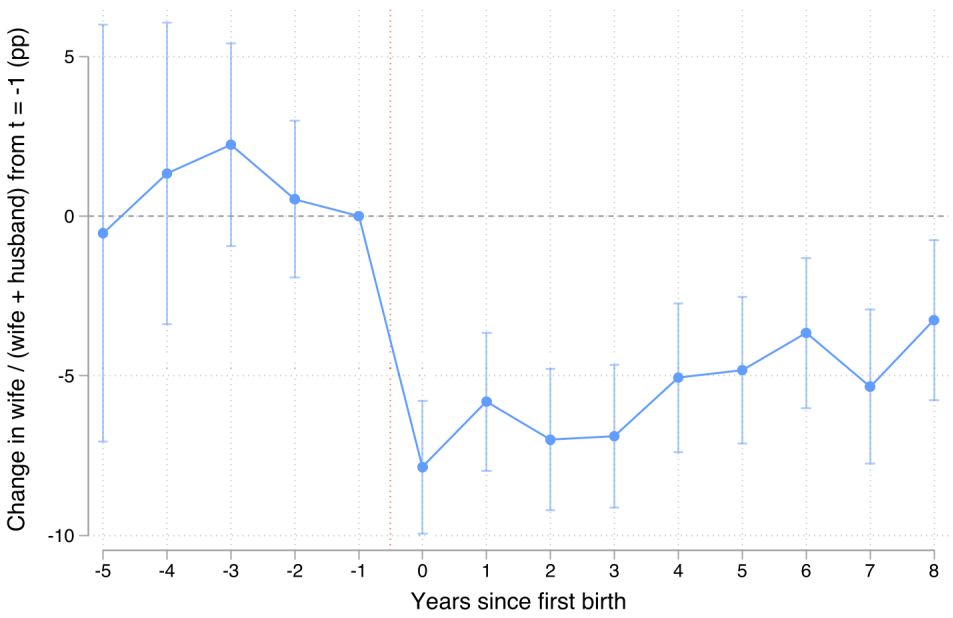


Figure 7: Wife's share of expenditures assigned to spouses

Notes: Coefficients from the woman fixed-effects event study of Figure 4 with the wife's share of spending assigned to either spouse, wife/(wife + husband), as the outcome, in percentage points relative to $t = -1$. Spending on children, other household members, and common expenses is excluded from the denominator.
Sample: Women whose first birth is observed in the panel, event times -5 to 8. Bars show 95 percent confidence intervals based on standard errors clustered by woman.
Source: JPSC.

sharply, while their time spent on housework and childcare increases substantially. Women's leisure time also declines, suggesting that the increase in household responsibilities is not accommodated solely through reductions in market work. In contrast, men's time allocation remains relatively stable across all three activities.

Taken together, these results show that the unequal changes in expenditures following childbirth are accompanied by similarly unequal adjustments in time use. Women experience declines in both individual expenditures and leisure time while taking on substantially more housework and childcare responsibilities. The JPSC allows us to document these changes in expenditures and time allocation within the same households, providing a more complete picture of how the economic consequences of childbirth differ between spouses.

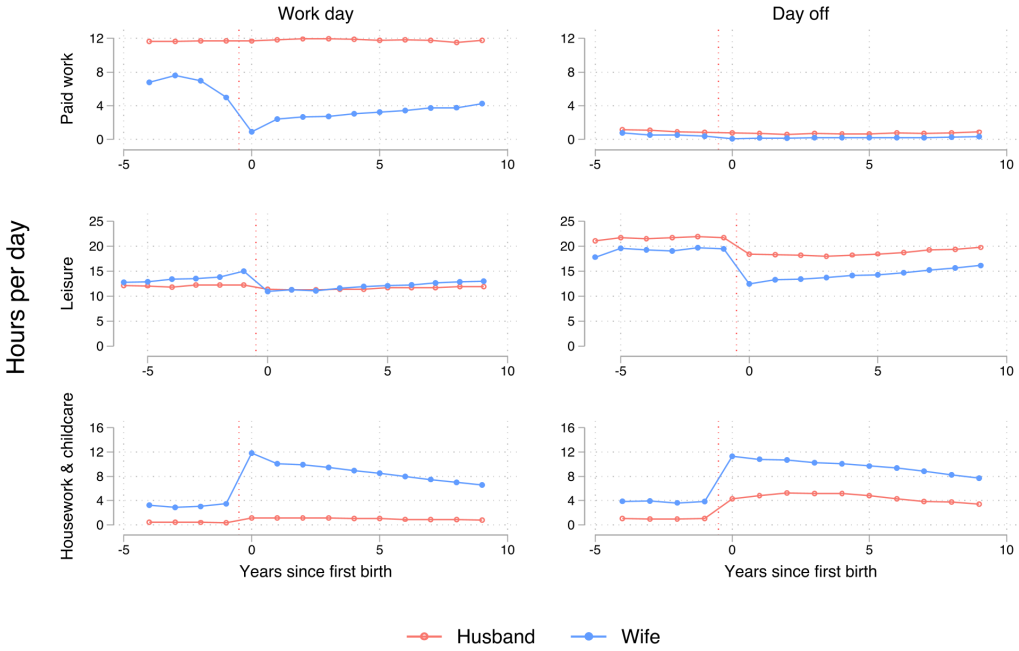


Figure 8: Time use around first childbirth

Notes: Mean hours per day by event time for wives and husbands on a workday (left) and a day off (right). Rows: paid work, including commuting and study; leisure, which includes hobbies and social activities as well as sleep, personal care, and meals; and housework and childcare.
Sample: Women whose first birth is observed in the panel, event times −4 to 9 (−6 to 9 for leisure), observations with both spouses' hours reported.
Source: JPSC.

5 Heterogeneity by Women's Labor-Market Attachment

The previous section documented a large and persistent decline in women's individual expenditures following childbirth. This section examines how these changes vary with women's labor-market attachment. We first consider differences in women's earnings shares before childbirth and then examine how expenditure and time-use patterns vary with employment status after childbirth.

5.1 Heterogeneity by Pre-Birth Earnings

We begin by examining whether changes in women's expenditure shares differ according to their economic position before childbirth. We divide women into quartiles based on their share of couple earnings at $t = -2$, two years before the first birth. Figure 9 plots the wife's

expenditure share for women in the bottom and top quartiles of this distribution.

Both groups experience a sharp decline in expenditure shares following childbirth. However, their trajectories diverge in the years after birth. By $t = 8$, women in the highest pre-birth earnings-share quartile have an expenditure share of approximately 7.6%, compared with 5.3% among women in the lowest quartile, a difference of 2.2 percentage points. These findings suggest that women's economic position before childbirth is associated with differences in the subsequent allocation of household expenditures.

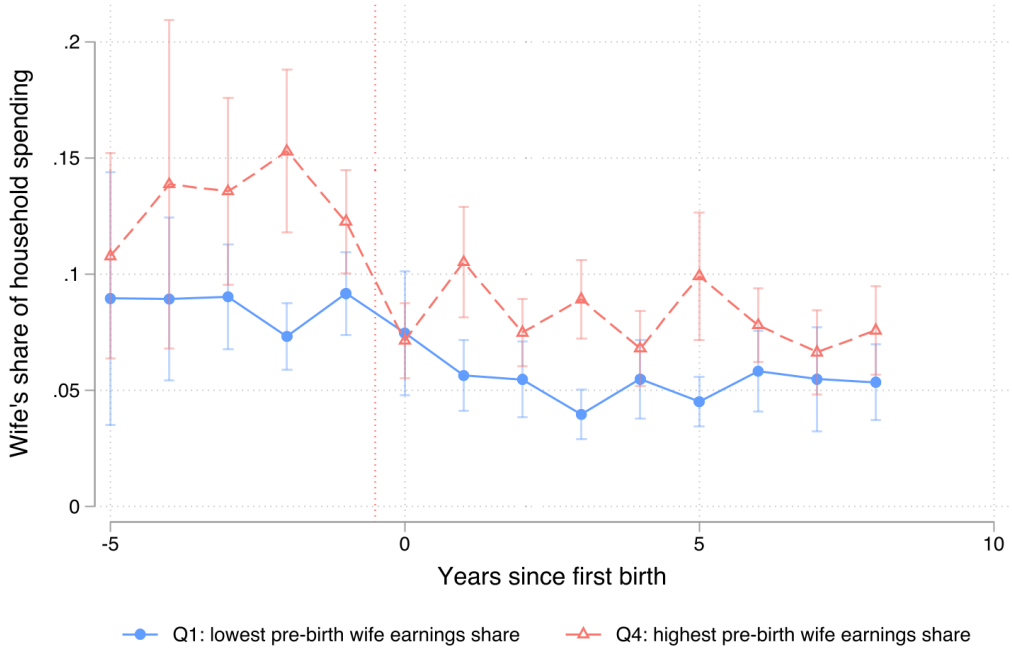


Figure 9: Wife's expenditure share by pre-birth earnings-share quartile

Notes: Mean wife expenditure share by event time, in share units, for women in the bottom (Q1) and top (Q4) quartiles of the wife's share of couple earnings, wife/(wife + husband), at $t = -2$. Quartiles are computed across women observed at $t = -2$. The expenditure share is spending assigned to the person divided by total household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses). Event times -5 to 8; cells with fewer than 10 observations are not shown. Bars show 95 percent confidence intervals of the means.
Source: JPSC.

5.2 Heterogeneity by Post-Birth Employment

We next examine how women's expenditure trajectories differ according to their employment status before and after childbirth. We classify women based on their employment in the two years preceding childbirth ($t = -2$ or $t = -1$) and five to eight years after childbirth

($t \in [5, 8]$). This yields four groups: (1) *recovered*, women who were employed both before and after childbirth; (2) *non-recovered*, women who were employed before childbirth but not in the post-birth window; (3) *late-attached*, women who were not employed before childbirth but were employed in the post-birth window; and (4) *never-attached*, women who were not employed in either period. Women whose first birth is observed but who have no observations in the post-birth window are classified as *censored*.

We use these categories to examine whether the persistence of the decline in women's expenditure shares varies with their subsequent employment trajectories. Because post-birth employment is itself an outcome of household decisions, these comparisons are descriptive and should not be interpreted as causal effects of employment.

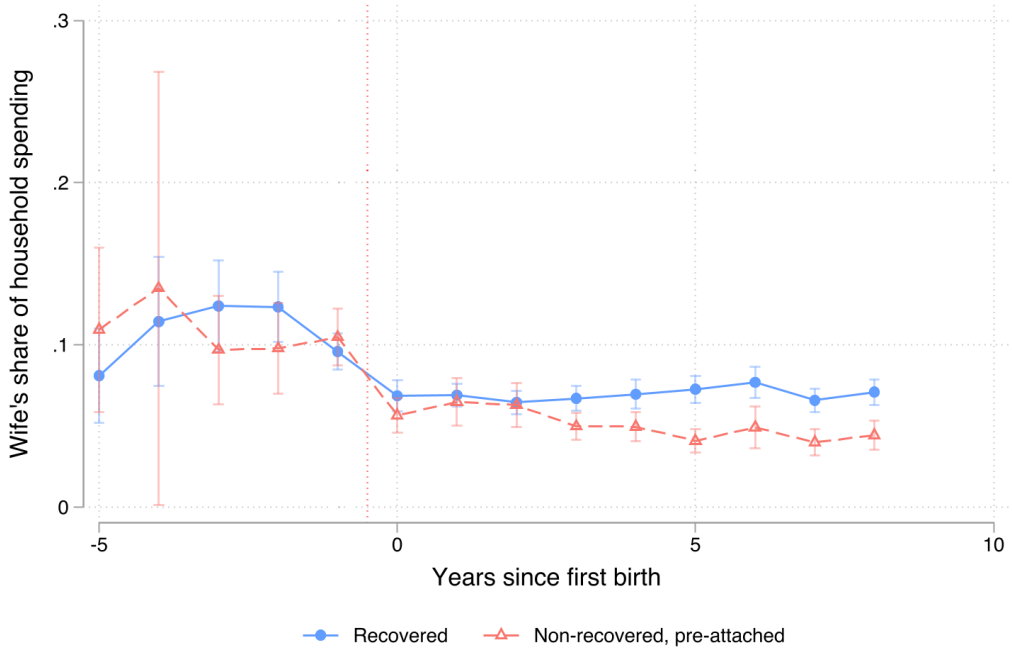


Figure 10: Wife's expenditure share by post-birth employment status

Notes: Mean wife expenditure share by event time, in share units. Recovered: employed at $t = -2$ or $t = -1$ and employed at least once in $t \in [5, 8]$. Non-recovered, pre-attached: employed at $t = -2$ or $t = -1$, observed in $t \in [5, 8]$, and not employed at any interview in that window. Employment is the wife's report that she is currently working. The expenditure share is spending assigned to the person divided by total household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses). Event times -5 to 8; cells with fewer than 10 observations are not shown. Bars show 95 percent confidence intervals of the means.
Source: JPSC.

These differences do not appear to be driven simply by differences in total household spending. Figure 11 shows that total household expenditures follow similar trajectories

across employment groups, increasing modestly following childbirth. Thus, the differences in women's expenditure shares are not accompanied by comparable differences in total household expenditures.

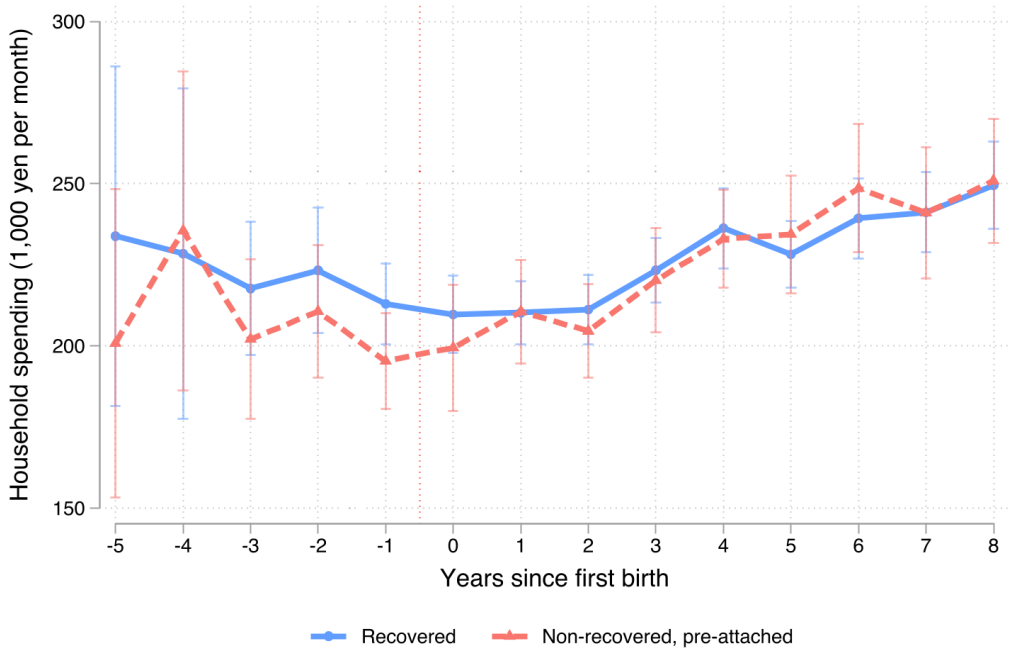


Figure 11: Total household expenditures by post-birth employment status

Notes: Mean total monthly household expenditure by event time, in thousands of real 2015 yen, for the two groups defined in Figure 10. Total expenditure is the sum of spending on the wife, the husband, children, other household members, and common expenses. Event times −5 to 8. Bars show 95 percent confidence intervals of the means.
Source: JPSC.

We also examine whether the expenditure differences between these groups were present before childbirth. At $t = -1$, the wife's expenditure share is 0.096 among recovered women, 0.105 among non-recovered women, 0.090 among late-attached women, and 0.097 among never-attached women. A joint F-test of equality across the four groups yields $F = 0.36$ ($p = 0.779$). These comparisons help assess the extent to which the differences observed after childbirth reflect divergence from pre-birth expenditure patterns.

Importantly, post-birth employment is not randomly assigned. Women who return to employment differ from those who do not, including in their pre-birth earnings. Appendix Table A2 shows that women with higher pre-birth earnings are more likely to return to work, while some differences also exist in husbands' education and earnings. We therefore interpret these results as descriptive associations between women's post-birth employment and their

expenditure shares, rather than causal effects of returning to work.

6 Potential Mechanisms

The previous sections document a large and persistent decline in women's individual expenditures following childbirth, with little corresponding change in men's expenditures. What explains this unequal adjustment? We consider two possible explanations. First, mothers may place greater value on their children's consumption than fathers do and consequently sacrifice more of their own consumption. Second, childbirth may weaken women's bargaining position within the household, leading to a reduction in their share of household resources. We develop a simple collective household framework to illustrate these mechanisms and examine the empirical evidence for each.

6.1 A Collective Household Framework

Consider a household consisting of a wife (w) and a husband (h). Following Chiappori (1992), we assume that household allocations are Pareto efficient, with weight μ on the wife's utility and $1-\mu$ on the husband's utility. Before the arrival of a child, the household solves

$$\max_{c_w, c_h} \quad \mu u(c_w) + (1-\mu)u(c_h) \qquad \text{s.t.} \qquad y = c_w + c_h, \tag{3}$$

where c_w and c_h denote individual consumption and y denotes total household resources.

Following childbirth, suppose both parents derive utility from their child's consumption, but potentially to different degrees. The household now solves

$$\max_{c_w, c_h, c_c} \quad \mu\left[(1-\alpha_w)u(c_w) + \alpha_w V(c_c)\right] \qquad +(1-\mu)\left[(1-\alpha_h)u(c_h) + \alpha_h V(c_c)\right] \tag{4}$$

$$\text{s.t.} \quad y = c_w + c_h + c_c, \tag{5}$$

where c_c denotes the child's consumption and α_w and α_h capture the weight that each parent places on the child's consumption relative to their own.

This framework illustrates two mechanisms through which childbirth may reduce the wife's consumption relative to her husband's.

First, mothers may be more altruistic toward their children than fathers ($\alpha_w > \alpha_h$).

In this case, the arrival of a child reduces the relative weight placed on the wife's own consumption, even if the spouses' Pareto weights remain unchanged. Mothers may therefore reduce their own consumption more than fathers to accommodate the child's needs.

Second, childbirth may change the relative Pareto weights of the spouses. In collective household models, Pareto weights may depend on factors that influence each spouse's position within the household, including their options outside marriage (Chiappori 1997; Lundberg and Pollak 1993). Childbirth could worsen women's outside options relative to men's, for example, through greater childcare responsibilities following divorce, reduced labor-market opportunities, or worse prospects of remarriage. A resulting decline in μ would reduce the relative weight placed on the wife's preferences and could lead to a decline in her share of household resources.

Both mechanisms can generate a disproportionate decline in women's consumption following childbirth. We next examine evidence that can help distinguish between them.

6.2 Evidence on Parental Altruism

We first examine whether differences in parental altruism can explain the observed expenditure patterns. If mothers place greater weight on their children's consumption than fathers do, they may sacrifice more of their own consumption to provide for their children. This explanation suggests that mothers may also bear a disproportionate share of the additional consumption needs associated with subsequent births. To investigate this possibility, we examine how individual expenditure shares change following the birth of the second child.

Figure 12 compares expenditure responses following the first and second births. While the first birth is associated with a substantially larger decline in the wife's expenditure share than the husband's, this asymmetry is not observed following the second birth. Instead, the expenditure shares of both spouses decline by similar amounts. This finding is difficult to reconcile with a simple explanation in which mothers disproportionately reduce their own consumption to accommodate each additional child. It suggests that the large gender difference in expenditure responses is particularly associated with the transition to parenthood, rather than recurring with every additional birth.

6.3 The Role of Outside Options

An alternative explanation is that childbirth changes women's relative bargaining position within the household. As illustrated by the collective household framework, changes in

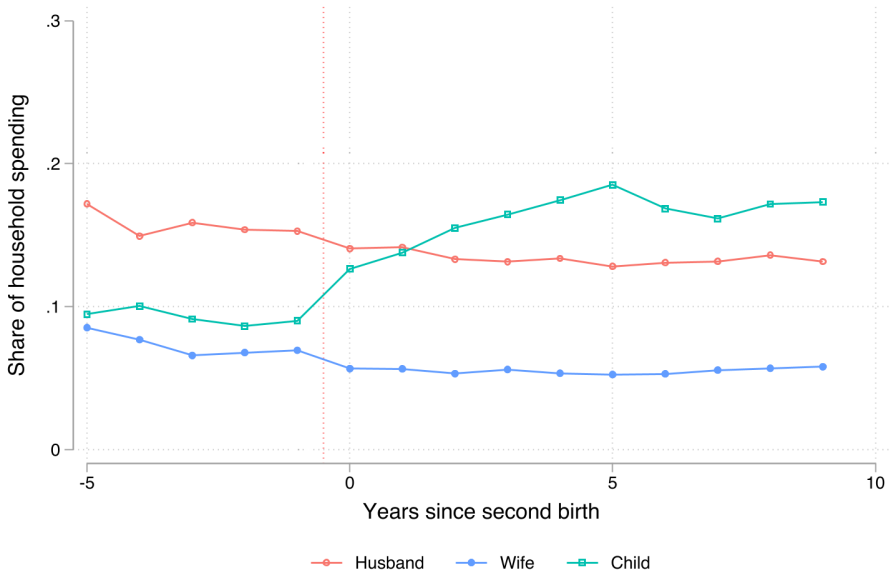


Figure 12: Changes in spouses' expenditure shares following the second birth

Notes: Mean expenditure shares of the husband, the wife, and children by event time relative to the second birth, in share units. The expenditure share is spending assigned to the person divided by total household expenditure (the sum of spending on the wife, the husband, children, other household members, and common expenses).
Sample: women whose second birth is observed in the panel, event times −5 to 9, observations with all three shares reported. Raw means; no controls.
Source: JPSC.

the wife's outside options may affect her Pareto weight and consequently, the allocation of household resources.

The heterogeneity results in Section 5 provide suggestive evidence consistent with this mechanism. Women with weaker labor-market attachment before childbirth, and those who do not return to employment afterward, tend to have lower expenditure shares in the years following childbirth. These patterns are consistent with the possibility that women's economic opportunities influence the allocation of household resources. However, labor supply and expenditure allocation are jointly determined, so these associations cannot establish a causal effect of employment on bargaining weights.

To examine the role of outside options more directly, we exploit a pension reform in Japan that improved women's financial position in the event of divorce without directly changing household resources during mariage. This provides a source of variation in women's outside options that allows us to examine whether strengthening these options mitigates the decline in women's expenditure shares following childbirth. The next section describes the reform

and our empirical strategy.

6.4 A Pension Reform Test of the Outside-Options Channel

To investigate whether women's outside options influence the allocation of household resources, we exploit a pension reform in Japan that improved women's financial security in the event of a divorce. Because the reform changed spouses' entitlements upon divorce without directly changing household resources during marriage, it provides a source of variation in outside options that can help us test this mechanism.

6.4.1 Institutional Background on the 2008 Pension Reform

Before the reform, employment-related pension entitlements were generally held in the name of the spouse who accrued them, typically the husband. This could leave women who had spent substantial periods outside the labor arket with relatively limited pension entitlements of their own in the event of divorce.

Japan introduced two provisions governing the division of pension entitlements upon divorce. Beginning in April 2007, divorcing spouses could divide employment-related pension entitlements accumulated during the marriage, up to an equal division, through mutual agreement or a court decision. In April 2008, an additional provision allowed economically dependent spouses of salaried workers to claim half of the pension records accrued during qualifying dependent periods from that date onward, without requiring their spouse's consent. These reforms potentially strengthened women's financial position outside marriage, particularly among women who depended on their husbands' earnings.[1]

Eligibility for the reforms depends on spouses' pension coverage. Salaried employees covered by the Employees' Pension Insurance system generally fall under Category 2, while their economically dependent spouses fall under Category 3. The 2007 provision potentially benefited couples with employment-related pension entitlements, whereas the 2008 provision provided additional rights specifically to Category 3 spouses.

6.4.2 Treatment and Control Groups

We use information on spouses' employment in the JPSC to identify women more likely to benefit from these changes. Our main treatment group consists of women whose husbands are

[1]See Toriyabe (2025) for the institutional background and a separate empirical analysis of the same reform on labor-supply and time-use margins among older households.

salaried employees, while the comparison group consists of women whose husbands are not covered by salaried employment. We also examine economically dependent wives separately, as they are more directly affected by the 2008 provision. We construct these groups using employment characteristics measured before the reform to avoid classifying households based on employment decisions that may themselves respond to the policy.

6.4.3 Effects on Intrahousehold Expenditure Allocation

We first examine whether the reform changed the allocation of expenditures between spouses generally, before investigating its implications for expenditure changes following childbirth.

We compare changes in the difference between wives' and husbands' expenditure shares before and after the reform among couples with salaried and nonsalaried husbands. To do so, we stack each couple's wife- and husband-assigned expenditure shares and estimate a triple-differences specification:

$$s_{ist} = \alpha_{is} + \lambda_t + \delta\left(W_s \times \text{Post}_t\right) + \beta\left(E_i \times \text{Post}_t\right) + \gamma\left(E_i \times W_s \times \text{Post}_t\right) + \varepsilon_{ist}, \qquad (6)$$

where s_{ist} is the expenditure share assigned to spouse s in couple i in year t, W_s is an indicator for the wife, and E_i identifies couples with salaried husbands before the reform. In this specification, $Post_t$ indicates years beginning in 2007, so the coefficient captures the differential change surrounding the introduction of the first pension-division provision. We include couple-by-spouse and calendar-year fixed effects and cluster standard errors by couple. The coefficient of interest, γ, measures whether the wife–husband expenditure-share gap changed differentially among couples more likely to benefit from the reform.

Table 1 presents the results. We find no statistically significant change in the allocation of expenditures toward wives in eligible households following the reform. The estimated difference is -0.26 percentage points (SE 1.49), and the result is similar when we allow for flexible year effects by spouse. We also find no clear change in women's employment following the reform.

These findings do not provide clear evidence that the reform changed expenditure allocation between spouses. However, the estimates are imprecise, partly because the sample of comparison households is relatively small. We therefore cannot rule out economically meaningful effects of the reform on intrahousehold allocation.

Table 1: Effects of the pension reform on intrahousehold expenditure allocation

	(1)	(2)
Eligible × Wife × Post (DDD)	-0.264	-0.263
	(1.490)	(1.490)
Eligible × Post	-1.505	-1.508
	(1.007)	(1.007)
Wife × Post	0.520	
	(1.459)	
Couple × spouse fixed effects	Yes	Yes
Calendar-year fixed effects	Yes	No
Wife- and husband-specific year fixed effects	No	Yes
Observations	9345	9345
Couples	1235	1235

Notes: The sample consists of couple-year observations from 2005–2008, with one row per spouse. The outcome is each spouse's share of total household expenditure, in percentage points. The sample is restricted to married women interviewed before 2007. Eligible equals one if the husband was a permanent employee at the wife's last interview before 2007. Post equals one for survey years 2007–2008. Column 1 includes couple-by-spouse and calendar-year fixed effects. Column 2 replaces calendar-year fixed effects with wife- and husband-specific year fixed effects, absorbing Wife × Post. Standard errors clustered at the couple level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. *Source:* JPSC.

6.4.4 Effects on Expenditure Allocation Following Childbirth

We next investigate whether the pension reform mitigated the decline in women's expenditure shares following childbirth. For each woman, we calculate the change in her expenditure share between the two years preceding her first birth and the birth year and following two years:

$$\Delta_i = \overline{s}^{w}_{i,[0,+2]} - \overline{s}^{w}_{i,[-2,-1]}, \tag{7}$$

We use this measure to compare expenditure changes around childbirth across birth cohorts with different exposure to the pension reform.

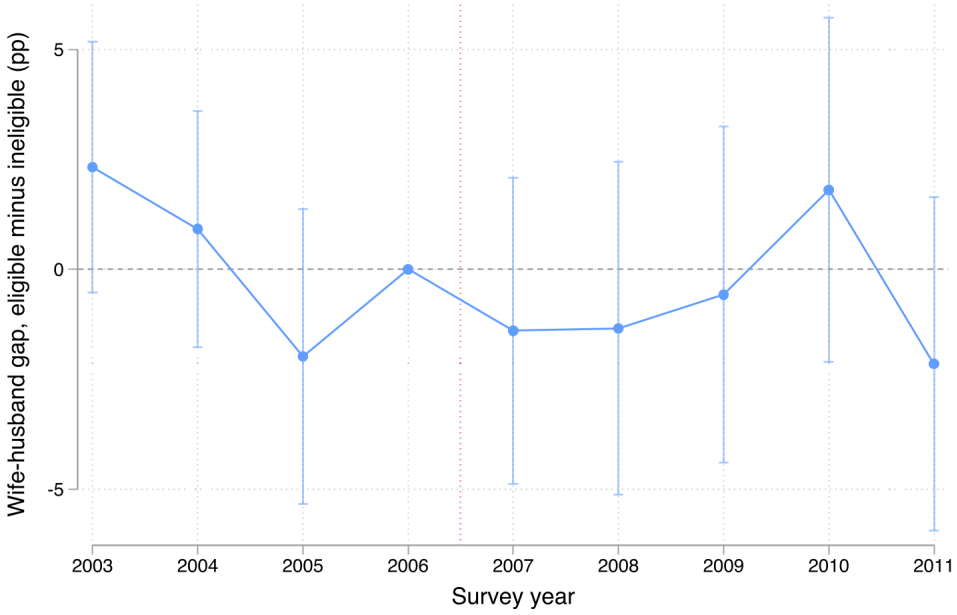


Figure 13: Evolution of the wife–husband expenditure-share gap around the pension reform

Notes: Coefficients on Wife × Eligible × year indicators, 2003–2011, from a regression of the stacked spouse expenditure shares (percentage points) with couple-by-spouse, spouse-by-year, and eligibility-by-year fixed effects; 2006 is the omitted year. Each coefficient is the change in the wife–husband share gap for eligible relative to ineligible couples, relative to 2006. Eligibility as in Table 1. The dotted line separates the 2006 and 2007 survey waves; the first pension-division provision took effect in April 2007. Bars show 95 percent confidence intervals based on standard errors clustered by couple.
Source: JPSC.

Comparison Across Broad Birth Cohorts. We first compare women who gave birth before and after the introduction of the reform. Our pre-reform group consists of women whose first child was born between 2000 and 2004, while the post-reform group consists of women whose first child was born between 2007 and 2011. We examine whether the change in expenditure shares around childbirth differs between these cohorts, and whether this difference is larger among women more likely to benefit from the pension reform.

We estimate

$$\Delta_i = \kappa_0 + \kappa_1 E_i + \kappa_2 C_i + \theta(E_i \times C_i) + u_i, \tag{8}$$

where E_i indicates greater reform exposure based on the husband's employment status measured before childbirth, and C_i indicates a first birth in 2007–2011. The coefficient θ captures the differential change in the childbirth-related expenditure penalty between the earlier and later birth cohorts among more-exposed women, relative to less-exposed women.

We find no statistically significant differential change in women's expenditure shares following childbirth. The estimated coefficient is -0.9 percentage points (SE 2.5, $p = 0.72$). Restricting attention to economically dependent wives, who were more directly affected by the 2008 provision, yields a similar estimate (-2.5 percentage points, SE 2.8).

As a placebo exercise, we repeat the comparison using two earlier birth cohorts, 1995–1999 and 2000–2004, whose childbirth windows precede the implementation of the reform. We find no statistically significant differential change in expenditure shares across these cohorts (1.3 percentage points, $p = 0.58$).

Interpretation. Overall, our preliminary results do not provide clear evidence that the pension reform changed intrahousehold expenditure allocation, either generally or following childbirth. However, the relatively small sample limits the precision of these estimates, particularly in specifications comparing women with different reform exposure. We therefore cannot draw strong conclusions about the magnitude of the effect of women's outside options on household expenditure allocation. In ongoing work, we are exploring additional datasets that would allow us to estimate these effects using a larger sample and more precisely assess the role of outside options in shaping the distribution of household resources.

7 Conclusion

The birth of a child has substantial consequences for women's employment and earnings, but much less is known about how these changes translate into the allocation of consumption within households. This paper examines how expenditures assigned to individual household members evolve following childbirth, using longitudinal data from Japan that distinguish expenditures on wives, husbands, and children from common household expenditures. These data allow us to study an important dimension of the economic consequences of childbirth that cannot be inferred from earnings alone.

The results reveal a large and persistent decline in women's individual expenditures following the birth of their first child. Women's expenditures fall to approximately half of their pre-birth levels and remain substantially lower even ten years after childbirth. In contrast, men's expenditures remain largely unchanged. Strikingly, total household expenditures also remain relatively stable despite the addition of a new household member. As expenditures on children increase, households appear to accommodate these new consumption needs largely through a reallocation of existing expenditures, with a disproportionate decline in spending

assigned to women. These findings demonstrate that the child penalty extends beyond labor-market outcomes to the allocation of expenditures within marriage. Importantly, the decline in women's expenditures does not mechanically follow from the decline in their earnings.

Childbirth is also accompanied by substantial changes in the allocation of time between spouses. Women's paid work hours and leisure decline, while their time spent on housework and childcare increases sharply. The corresponding changes for men are considerably smaller. Moreover, the persistence of women's expenditure losses varies with their labor-market attachment. Women with higher earnings shares before childbirth and those who return to employment afterward tend to have higher expenditure shares in the years following childbirth. Husbands also contribute more to household work on weekdays when their wives return to employment. Although these relationships are descriptive, they show that the allocation of expenditures, market work, and household responsibilities are closely related.

To understand the mechanisms underlying these patterns, we consider two explanations within a collective household framework. The first is that mothers place greater weight on their children's consumption than fathers do and consequently sacrifice more of their own consumption. The second is that childbirth weakens women's outside options, potentially reducing their relative Pareto weights within the household. The pronounced gender difference in expenditure responses following the first birth is not repeated following the second birth. This pattern is difficult to reconcile with a simple explanation in which mothers disproportionately sacrifice their own consumption for each additional child, although it does not rule out differences in parental altruism more generally.

To investigate the role of outside options, the analysis exploits a pension reform in Japan that improved certain women's financial entitlements in the event of divorce without directly changing household resources during marriage. The analysis examines whether the reform affected expenditure allocation between spouses generally and whether it mitigated the decline in women's expenditure shares following childbirth. Preliminary estimates do not reveal a clear change in expenditure allocation following the reform. However, the estimates are imprecise, and economically meaningful effects cannot be ruled out. In ongoing work, we are exploring additional datasets that would allow us to investigate this mechanism with greater statistical precision.

Overall, the findings highlight the importance of looking beyond earnings to understand the economic consequences of childbirth for women. The substantial decline in mothers' individual expenditures, alongside the relative stability of fathers' expenditures, raises important

questions about how couples share the costs of raising children. Distinguishing whether these patterns reflect differences in parental preferences or changes in spouses' relative influence over household decisions is important for understanding women's economic well-being following childbirth. If changes in outside options contribute to the observed expenditure patterns, policies that strengthen women's economic opportunities and financial security outside marriage may have consequences for how resources are allocated within marriage, beyond their direct effects on women's earnings. Establishing the importance of this channel remains an important direction for future research.

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A Appendix

A.1 Raw expenditure-share and earnings-share trajectories

Figure A1 plots the raw wife earnings-share trajectory around the first birth. Figure A2 decomposes the household budget into wife-, husband-, and child-attributed shares. The plotted expenditure shares do not sum to one. The residual, the common pool plus spending on other household members, absorbs most of the household budget, averaging 65% of total spending after birth.

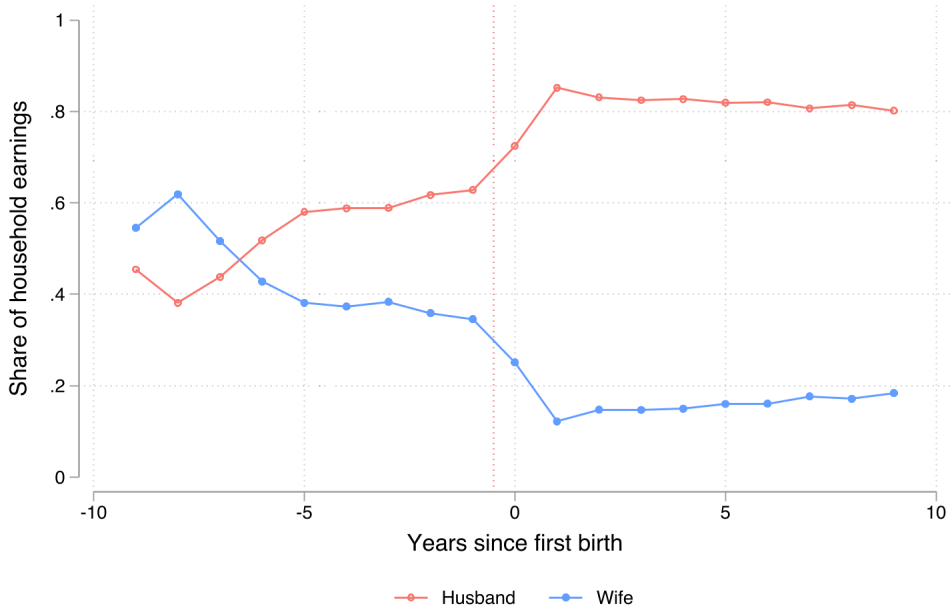


Figure A1: The wife's share of couple earnings collapses at first birth and stays low

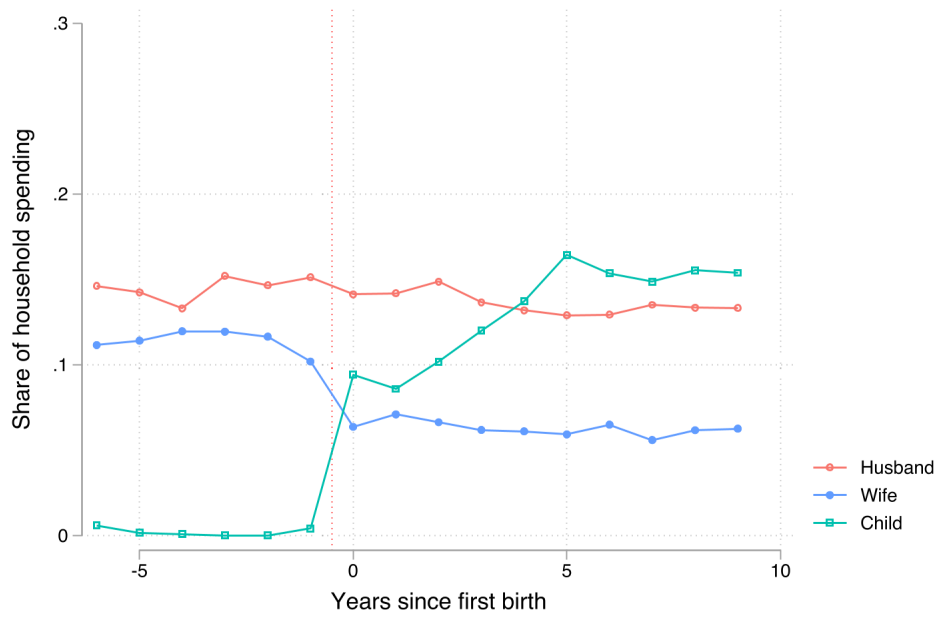


Figure A2: The child's expenditure share rises as the wife's falls, while the husband's is roughly flat

A.2 Cohort composition and identifying variation

Figure A3 plots the raw wife expenditure share by event time separately for each JPSC recruitment cohort. The post-birth decline appears within every cohort with adequate cell sizes, so the pooled fact is not an artifact of which cohort contributes which event times. This within-cohort pattern is also the cohort-aggregate version of the headline, complementing the woman-level estimates. Long-horizon persistence (t beyond 8) necessarily draws on the earlier cohorts only, which is why the main figures stop at $t = 8$. Figure A4 shows the distribution of the wife's age at first birth by cohort. This variation separates age from event time in the KLS specification of Section 4.1.

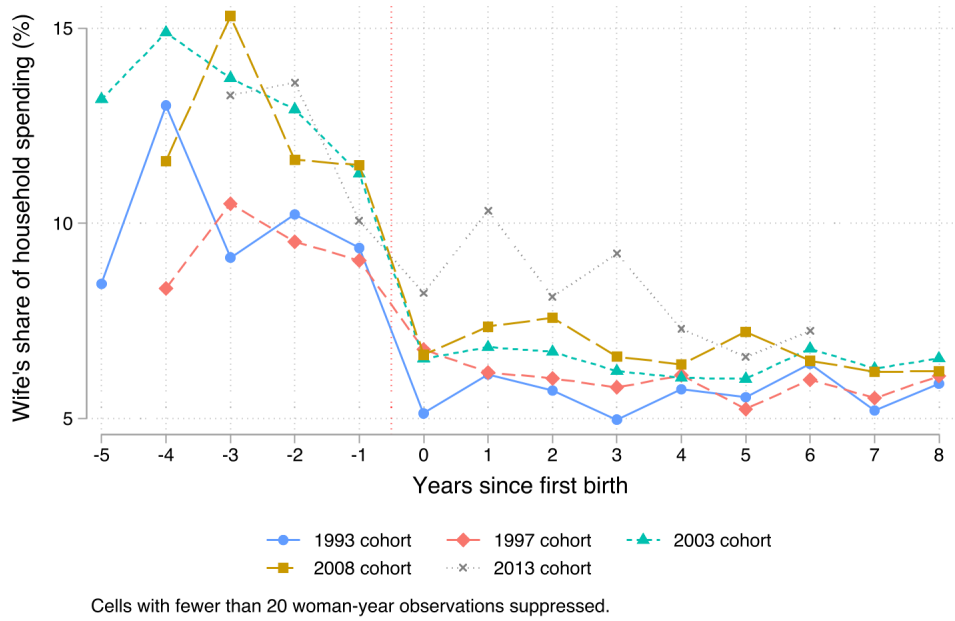


Figure A3: The wife-share decline appears within every JPSC recruitment cohort

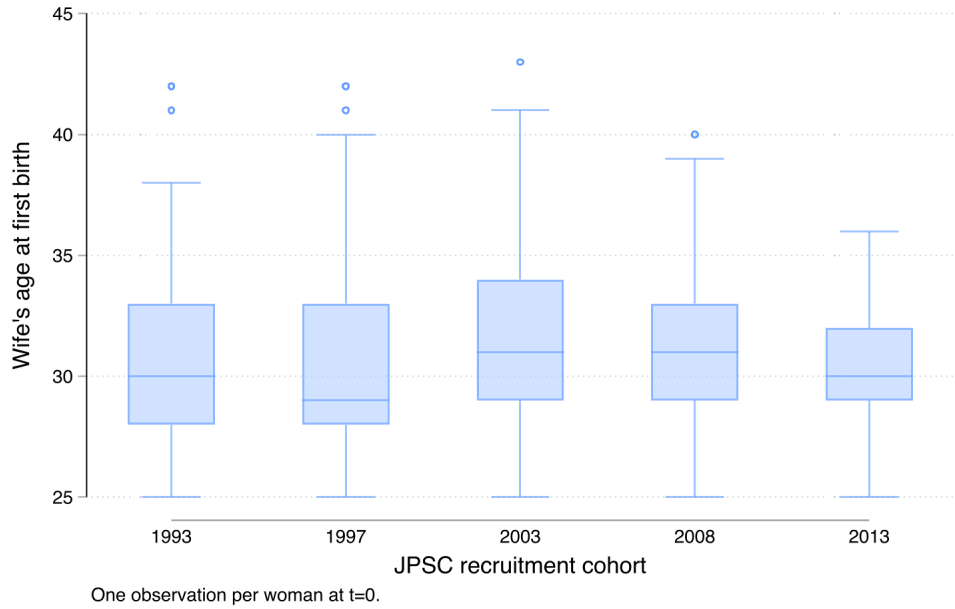


Figure A4: Age at first birth varies widely within every entry cohort, the variation identifying the KLS specification

A.3 Spouse-stacked compact pre/post regression

Table A1 reports the compact pre/post collapsed spec backing the Section 1 prose claim about the wife-vs-husband post-birth contrast. The wife's share drops 3.4 percentage points post-birth and the husband's drops 0.9 percentage points, so the wife's share drops 2.5 percentage points more than the husband's (coefficient +0.025, SE 0.004, $N = 15387$, 882 clusters).

Table A1: The wife's post-birth share drop exceeds the husband's by +0.025 pp in the spouse-stacked specification

	(1) b/se
Post	-0.035
	(0.004)
Husband-row × Post	0.025
	(0.004)
Observations	15387
Women (clusters)	882

Notes: Spouse-stacked panel: each woman contributes two rows per event-time (wife-attributed share and husband-attributed share). The coefficient *Post* is the wife-side post-birth change (since the husband-row indicator H has $H = 0$ for the wife row); *Husband-row* × *Post* is the husband-row differential, i.e., the wife-vs-husband post-birth contrast. Fixed effects: woman-by-spouse-role and calendar year. Standard errors clustered at the woman level.

A.4 Recovery-group balance on pre-birth observables

Table A2 reports group means for the four recovery groups defined in Section 5. Pre-birth earnings differ across groups mechanically, since late- and never-attached women were not working at $t = -2$, but also in level among the pre-attached: recovered women earned more before birth than non-recovered pre-attached women, and a logit of recovery on pre-birth earnings, education, and ages has meaningful classifying power. This is why we treat realized recovery as partly predicted by pre-birth economic position rather than as a source of independent variation. Husband education and earnings also differ across groups.

Table A2: Recovery groups differ on pre-birth position: recovered women earned more before birth

	1	2	3	4	Total
Marriage age	27.27	26.77	26.00	27.60	27.06
Wife education (1=Jr high, 7=Grad)	4.79	4.73	4.27	4.60	4.69
Husband age at first birth	32.85	32.63	31.00	34.70	32.94
Husband education (1=Jr high, 7=Grad)	4.83	5.67	3.64	4.90	4.95
Wife earnings at t=-2 (1000 yen)	295.34	196.19	55.06	79.25	204.15
Husband earnings at t=-2 (1000 yen)	387.21	460.18	441.29	510.22	435.32

Notes: Columns 1–4 are the four recovery groups (1 = Recovered; 2 = Non-recovered, pre-attached; 3 = Late-attached; 4 = Never attached). Column "Total" pools the four. Earnings in 1,000 yen, 2015 real. Education codes on 1–7 scale.

A.5 A second pre-determined margin: regular employment

The pre-determined margin that carries the clearest gradient is the pre-birth earnings share in the main text (Figure 9). A second pre-birth margin points the same way. Regular employment is a status tied to statutory parental-leave rights and re-entry in Japan. Splitting on pre-birth regular versus non-regular employment produces a modest separation of 1.0 percentage points in the post-birth wife share (Figure A5). Both margins are fixed before the birth, so neither is an outcome of the post-birth allocation.

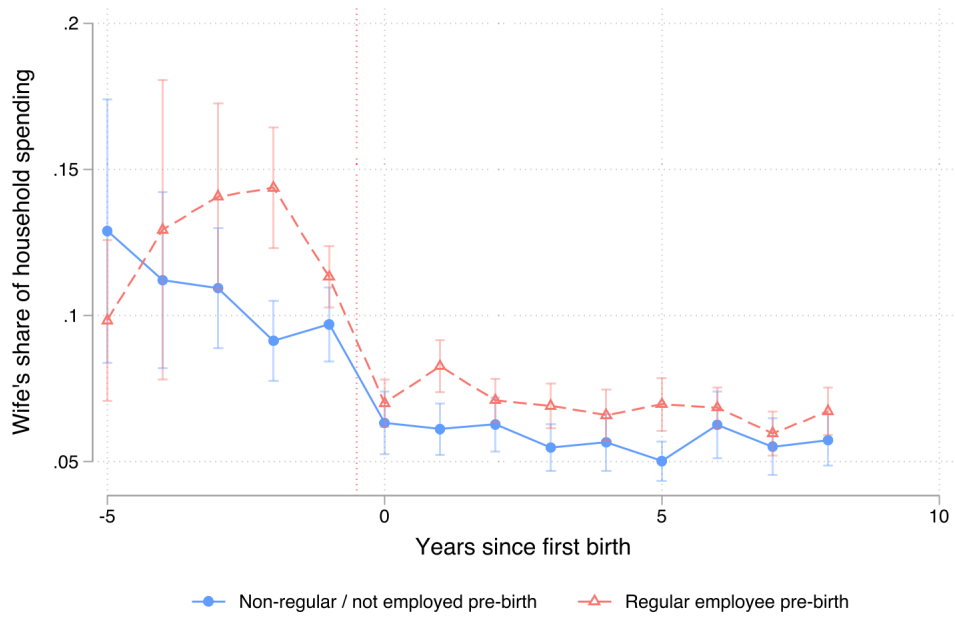


Figure A5: Pre-birth regular employment is associated with a modestly higher post-birth wife share

A.6 Four-group recovery trajectories

The main-text recovery figures use the two-group contrast (recovered vs. non-recovered pre-attached) that carries the paper's narrative. Figures A6 and A7 report the four-group versions including the late-attached and never-attached groups.

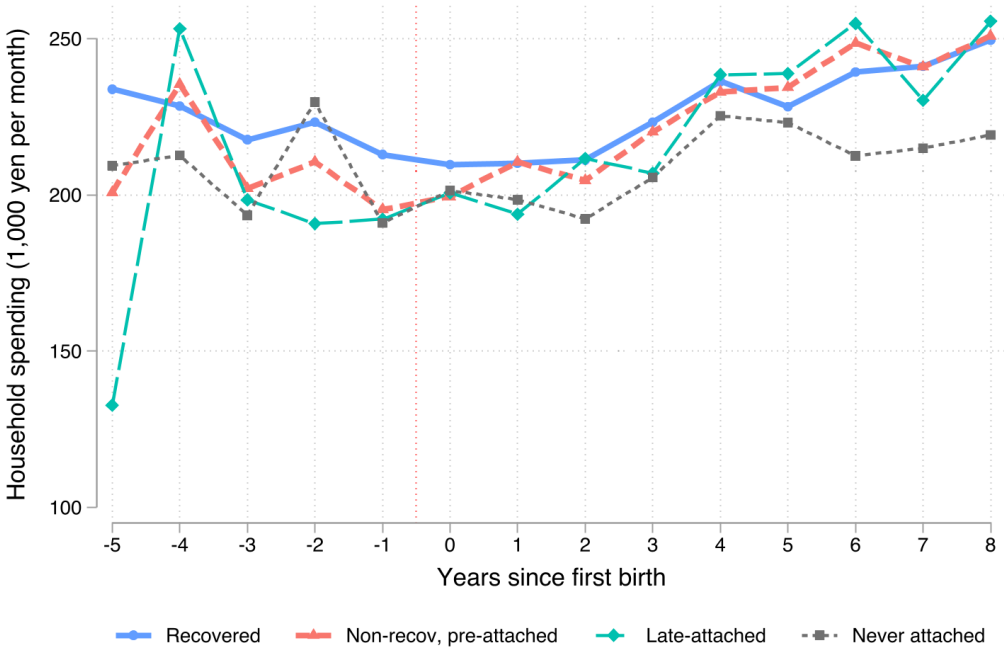


Figure A6: Total household expenditure by all four recovery groups

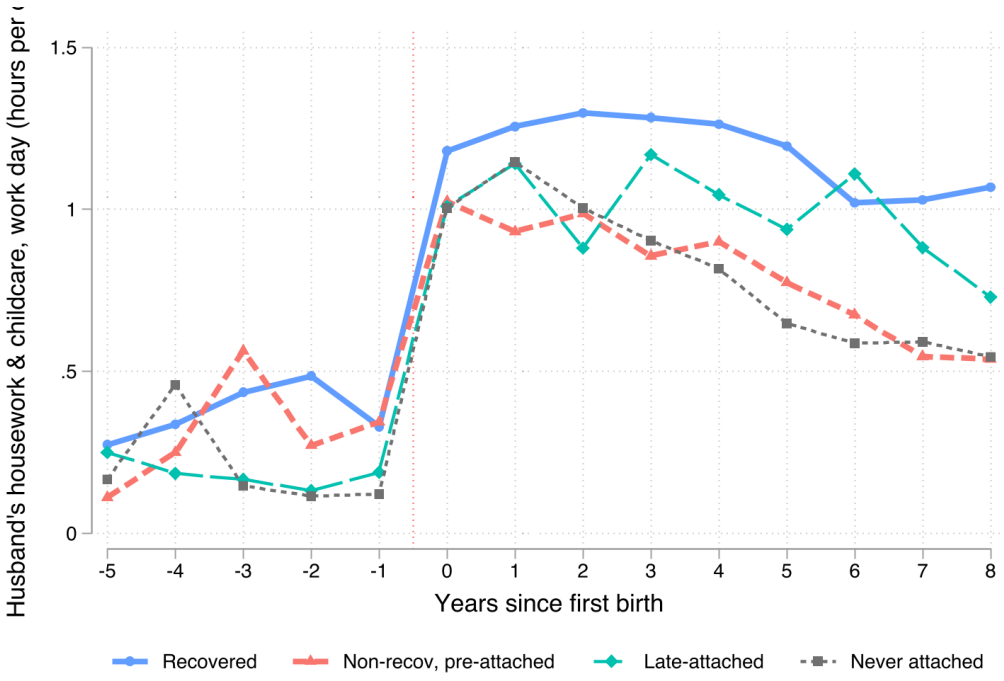


Figure A7: Husband weekday housework by all four recovery groups

A.7 Denominator and pocket-money checks

The main allocation object uses the full household budget as the denominator, because the child is a new claim on household resources. Two appendix checks show how the result looks under narrower objects. First, Figure A8 removes the child bucket and common pool and asks whether the recovery-group contrast remains among attributed adult spending only. Second, Figures A9–A10 use the separate *kozukai* pocket-money item as a measure of husband-controlled cash. These checks support the main decomposition, but they do not replace it.

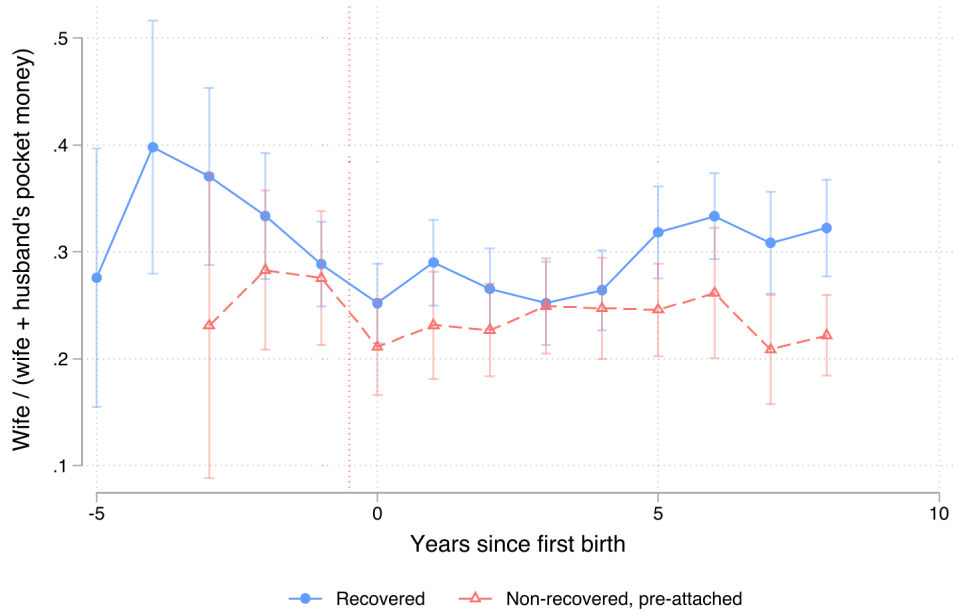


Figure A8: Adult-only denominator check: the recovery-group contrast remains after removing the child bucket

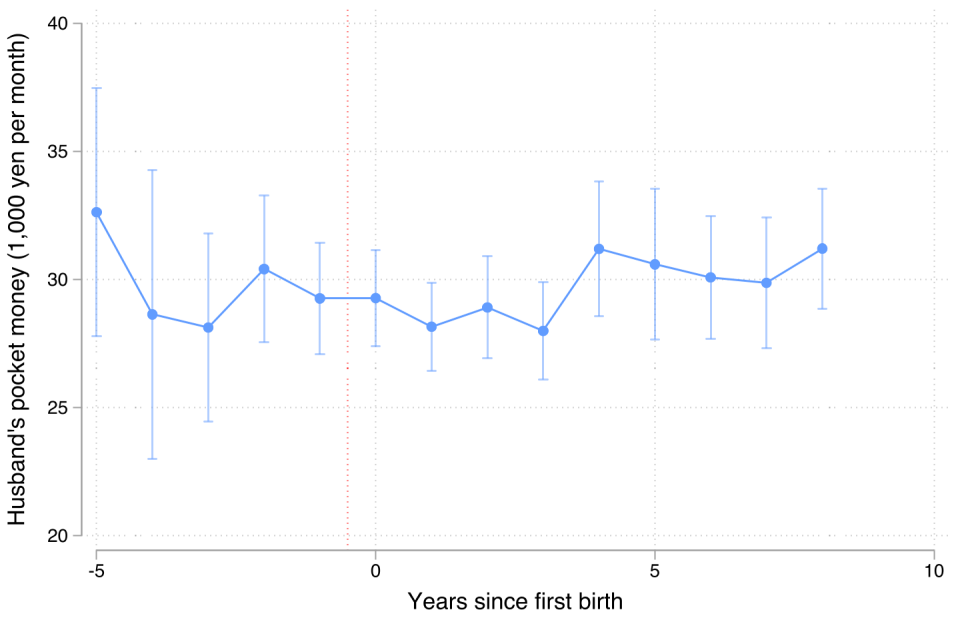


Figure A9: Appendix check: the husband's pocket money is flat in yen through the first birth

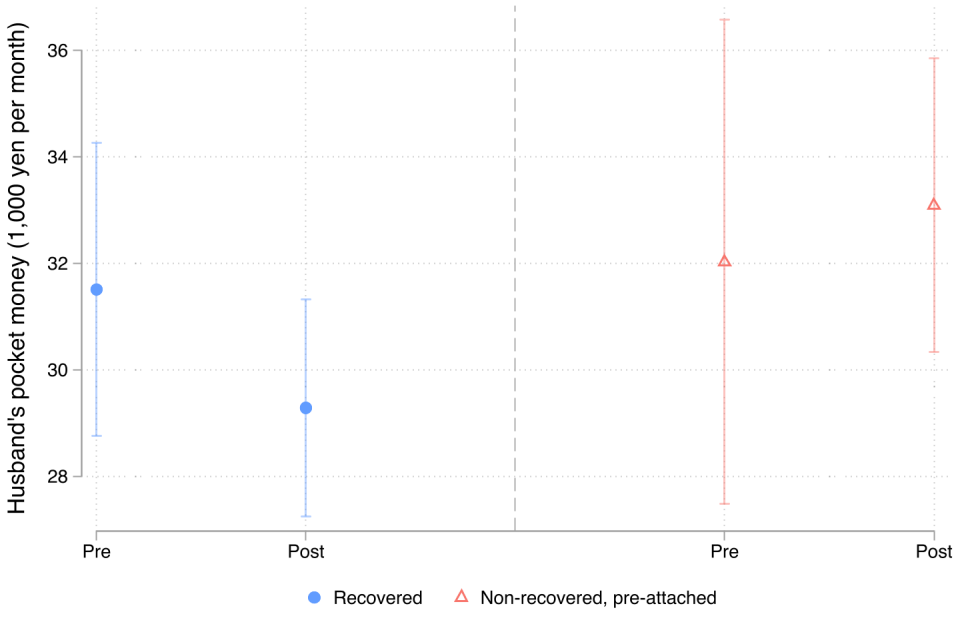


Figure A10: Appendix check: husband pocket money shows no recovery-group gradient

A.8 Common expenses and other household members

The common pool falls modestly at birth, from about 70 percent of household expenditure in the pre-pregnancy period to around 65 percent in the five years after childbirth (Figure A11). Household composition is stable through the same window. Fewer than 0.5 percent of women report that either set of grandparents moves in after the birth, and the flat trajectory of spending on other household members in Figure A11 corroborates this. The mechanism analysis in Section 6 therefore does not need to contend with coresidence changes as a confound.

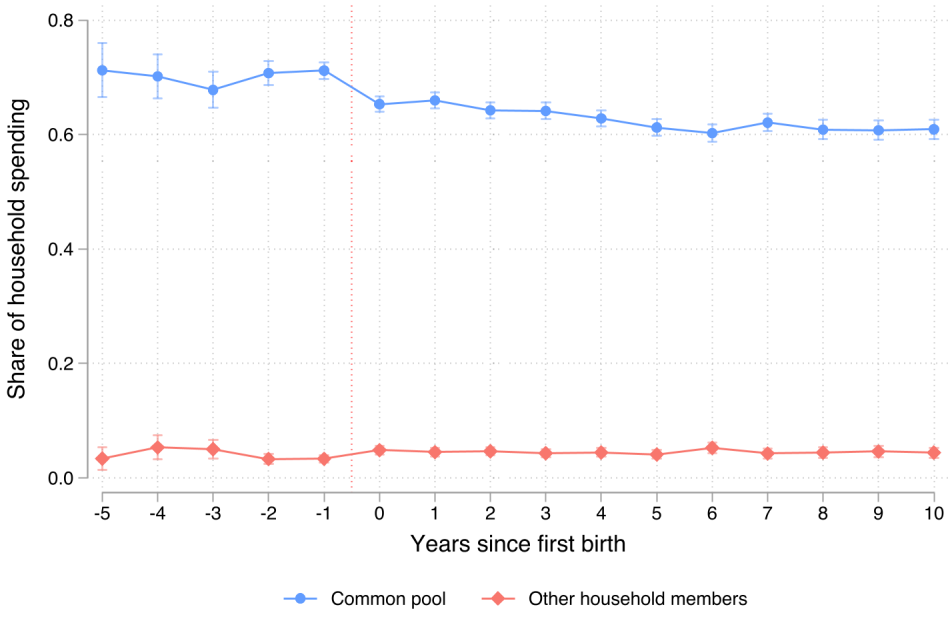


Figure A11: The common pool falls at birth while spending on other members is flat

A.9 Pension-reform designs rejected along the way

This appendix documents two birth-anchored designs that Section 6.4 rejects, and why. Reporting them matters. A referee who reaches for the birth clock should see that we tried it, and small-cell birth-cohort contrasts in this panel produce estimates that look like effects and are not.

The first is the narrow within-woman birth-change DiD, estimated with the Category-3 intensity definition. On birth cohorts 2003–2004 versus 2007–2008, the eligible-ineligible difference in how the first birth changes the wife's share is +11.4 percentage points (SE

4.4, $p = 0.01$). The estimate rests on 11 eligible post-reform mothers. Its placebo fails: the same contrast across two pre-reform cohort pairs (first births 2003–2004 versus 1999–2000, no reform between them) gives -9.0 percentage points ($p = 0.00$). A large, significant estimate where no reform exists means the eligible-ineligible cohort gap is unstable at this width. Widening the cohorts to 2000–2004 versus 2007–2011 removes both problems, and that validated design is the one Section 6.4 reports.

The second is an event-time DDD. The treated cohort consists of first births in 2010–2012, all post-reform; the control cohort consists of first births in 1999–2001, all pre-reform. Both cohorts are differenced through the birth transition on a common event clock. Eligibility is measured as Category-3 dependent-spouse status at $t = -3$ and carried within woman. The coefficient of interest is the triple interaction Eligible $\times$ Treated $\times$ Post in a regression for the wife's expenditure share. Under the bargaining hypothesis, eligible wives in the post-reform cohort should retain a higher expenditure share after birth than comparable ineligible wives, relative to the same eligible-ineligible contrast in the pre-reform cohort.

Table A3 reports the result. The triple coefficient is -0.014 (SE 0.041, $p = 0.73$; $N = 419$, 63 women). A pre-policy placebo comparing 1996–1998 to 2001–2003 yields -0.032 ($p = 0.43$). A stacked levels variant gives -7.4 thousand yen per month ($p = 0.41$) for the wife–husband level differential. Stacked-share four-way variants are also null: -0.026 with woman-by-spouse fixed effects and -0.035 with woman-by-year fixed effects.

Table A3: The pension-reform DDD is null and underpowered: no detectable eligibility effect on the wife's share

	Wife share		Levels	Stacked differences	
	(1)	(2)	(3)	(4)	(5)
	Main	Placebo	1,000 yen	Share	Share
Eligibility effect	-0.014	-0.032	-7.4	-0.026	-0.035
	(0.041)	(0.041)	(8.9)	(0.052)	(0.051)
Woman FE	Yes	Yes	No	No	No
Woman × spouse FE	No	No	No	Yes	No
Woman × year FE	No	No	Yes	No	Yes
Observations	419	345	840	838	838
Women	63	52	63	63	63

Notes: Standard errors, clustered by woman, are in parentheses. No coefficient is statistically significant at conventional levels; with the treated-eligible cell sizes reported in the text, the design is underpowered against economically relevant effect sizes. The real-cohort comparison is first births in 1999–2001 versus 2010–2012; the placebo comparison is 1996–1998 versus 2001–2003. Eligibility is Category-3 dependent-spouse status measured at $t = -3$. Columns 1–2 estimate Eligible × Treated × Post on the wife's expenditure share. Columns 3–5 use stacked wife–husband rows and estimate Wife × Eligible × Treated × Post; column 3 uses monthly expenditure levels in 1,000 real 2015 yen, while columns 4–5 use expenditure shares. $^{*}\,p < 0.10$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

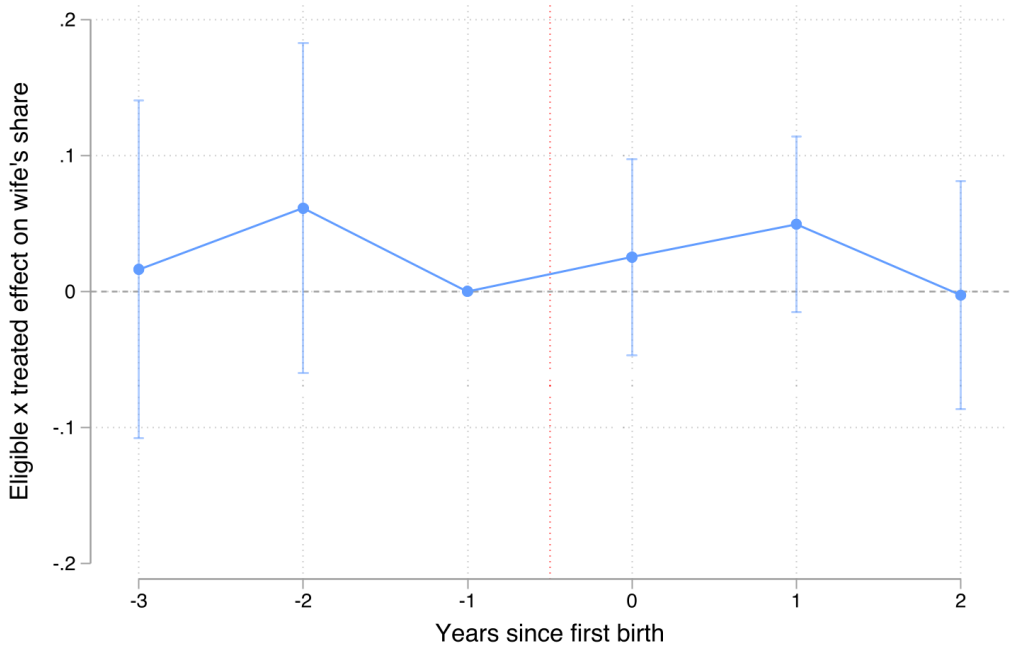


Figure A12: The two-cohort pension-reform design does not isolate a post-birth wife-share effect

Figure A12 plots the woman-FE event-time analogue of column 1, normalized to $t = -1$. The saturated event-time specification mechanically drops the final $t = +3$ bin, so the figure reports the cleanly estimated window from $t = -3$ through $t = +2$.

This is a careful attempt, but not a clean identification design. The reform is a calendar-time outside-option shock, not a birth-cohort treatment, so the policy clock and the childbirth clock do not separate cleanly on the event-time margin. Pre-reform birth cohorts become exposed in their post-2007 event years; for late births, eligibility measured before birth is not necessarily pre-policy; and the treated-eligible cells are small. This is why the main text tests the reform on the calendar clock, where the policy timing is unambiguous, and on validated wide birth cohorts, where the placebo passes. Both give precise nulls.